MPG No. 31.1.115 $\bar{4}3m$ [Type I, cubic] [Q tensor]

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & 0 \\ 0 & Q_{xx} & 0 \\ 0 & 0 & Q_{xx} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ Q_{yzx} & 0 & 0 \\ 0 & Q_{yzx} & 0 \\ 0 & 0 & Q_{yzx} \end{bmatrix}$$

$$Q_{yzx} = \mathbb{Q}_{xyz}^{(1)}$$

* Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxyy} & 0 & 0 & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxyy} & 0 & 0 & 0 \\ Q_{xxyy} & Q_{xxyy} & Q_{xxxx} & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{yzyz} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_4^{(1)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_4^{(1)}$$
$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_4^{(1)}$$

* Rank 4 tensor (ssa). * Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & 0 \\ 0 & Q_{yzyz} & 0 \\ 0 & 0 & Q_{yzyz} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(3)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 \\ Q_{xxyy} & Q_{xxyy} & Q_{xxxx} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzyz} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_4^{(1)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_4^{(1)}$$
$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_4^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & -Q_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & -Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & -Q_{yzyz} \end{bmatrix}$$

$Q_{yzyz} = \mathbb{Q}_0^{(3)}$

* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & 0 & 0 \\ 0 & Q_{xxxx} & 0 \\ 0 & 0 & Q_{xxxx} \\ 0 & 0 & Q_{yyzz} \\ Q_{yyzz} & 0 & 0 \\ 0 & Q_{yyzz} & 0 \\ 0 & Q_{yyzz} & 0 \\ 0 & 0 & Q_{yyzz} \\ Q_{yyzz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$Q_{xxxx} = 3\mathbb{Q}_0^{(1)} + 2\mathbb{Q}_4^{(1)}$
$Q_{yyzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_4^{(1)}$